

Supplementary Tables: Multispectral Indices Results on SeasonNet Dataset for All Seasons (Fall, Snow, Winter, Spring, Summer)

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Description: This supplementary file contains the multispectral indices results per season for the SeasonNet dataset.

Supplementary Table S11: Pixel level segmentation results on the SeasonNet dataset for the fall images. Each row represents one multispectral index. Accuracy, precision, recall, F1 score, and IoU are reported in percent. TP, TN, FP, and FN are total pixel counts summed over the complete fall split.

No.	Index	Acc (%)	Prec (%)	Rec (%)	TP	TN	FP	FN	F1 score (%)	IoU (%)
1	VrNIR-BI	84.32	87.35	79.35	82212377	98348227	11911145	21659156	83.04	71.01
2	VgNIR-BI	83.70	86.72	80.15	79274302	83855798	12136685	19632430	83.31	71.39
3	BBI	84.47	93.55	77.00	314099	313965	21673	93820	84.47	73.12
4	CBCI	82.46	87.33	79.69	121814706	107269838	17671262	31051960	83.33	71.43
5	BAEI	85.34	88.82	77.97	86459198	119164495	10885295	24430004	83.04	71.00
6	BCI	83.86	87.17	81.54	114669299	107936986	16871842	25968183	84.26	72.80
7	BLFEI	83.93	87.95	80.01	115636155	117894142	15841950	28886057	83.79	72.11
8	BRBA	86.30	88.86	77.45	67101509	109011160	8409053	19538222	82.76	70.60
9	BUb	80.54	85.80	80.21	63139299	44574684	10451864	15574600	82.91	70.81
10	BUc	81.46	87.72	78.37	48198003	39878510	6746931	13302455	82.78	70.62
11	BUI	84.20	88.22	79.61	111516054	119907735	14885286	28556834	83.70	71.97
12	CBI	86.64	87.90	77.37	29625468	52967725	4078150	8663260	82.30	69.93
13	ENDISI	81.59	87.80	77.88	71444454	62424846	9925048	20287913	82.55	70.28
14a	IBI with NDVI	79.98	84.30	82.56	92538721	54448412	17239495	19554468	83.42	71.55
14b	IBI with SAVI	80.54	84.57	82.13	87440924	57329882	15950861	19029564	83.33	71.43
15	MBAI	81.67	86.52	78.18	53994822	50625034	8415931	15068865	82.14	69.69
16	NBAI	84.36	86.78	80.29	57268080	65654022	8724420	14062400	83.41	71.54
17	NBI	81.51	83.37	71.42	55446560	91138758	11057314	22189902	76.93	62.51
18	NDBI	80.17	83.83	83.82	100350128	56209318	19356521	19370454	83.83	72.15
19	PISI	82.89	87.92	81.48	36509084	27974679	5015531	8296527	84.58	73.28
20	RNDSI	80.52	85.76	80.42	89076514	61775010	14793857	21690773	83.00	70.94
21	SwiRed	74.55	87.49	74.59	14829755	6174195	2119946	5050848	80.53	67.41
22	UBI	80.22	85.70	82.29	103922465	57114045	17338513	22362271	83.96	72.36
23	UI	81.28	85.57	82.17	113431917	76448354	19124318	24621373	83.83	72.17
24	VIBI	75.84	85.04	80.56	40996024	12683760	7209689	9889848	82.74	70.57
25	WE-NDBI	82.34	85.81	84.08	173310693	113325293	28648359	32826234	84.94	73.82

Supplementary Table S12: Pixel level segmentation results on the SeasonNet dataset for the snow images. Each row represents one multispectral index. Accuracy, precision, recall, F1 score, and IoU are reported in percent. TP, TN, FP, and FN are total pixel counts summed over the complete snow split.

No.	Index	Acc (%)	Prec (%)	Rec (%)	TP	TN	FP	FN	F1 score (%)	IoU (%)
1	VrNIR-BI	82.44	83.70	87.07	3573091	2184156	695889	530631	85.35	74.45
2	VgNIR-BI	82.94	83.53	87.87	3525882	2222453	695454	486924	85.64	74.89
3	BBI	86.01	85.14	94.00	3094818	1441360	540210	197698	89.35	80.75
4	CBCI	81.52	82.91	87.43	2743073	1491337	565285	394318	85.11	74.08
5	BAEI	89.27	85.59	85.25	597426	1098558	100590	103351	85.42	74.55
6	BCI	85.10	82.95	84.70	1306276	1574798	268550	235961	83.81	72.14
7	BLFEI	80.49	79.99	86.36	799263	545941	199941	126190	83.06	71.02
8	BRBA	86.73	84.77	80.31	200092	355257	35936	49058	82.48	70.19
9	BUb	84.87	86.59	82.52	2891501	3054818	447732	612327	84.51	73.17
10	BUc	87.33	86.44	88.59	4854684	4700068	761475	625225	87.50	77.78
11	BUI	88.96	83.96	85.28	760422	1468561	145326	131274	84.61	73.33
12	CBI	86.75	86.06	84.85	2845168	3496463	460884	507820	85.45	74.60
13	ENDISI	82.81	83.79	83.84	2083827	1794299	403050	401742	83.81	72.14
14a	IBI with NDVI	83.30	86.49	88.47	5514162	2365955	861302	718686	87.47	77.73
14b	IBI with SAVI	83.51	86.67	88.51	5583700	2433284	858601	724726	87.58	77.91
15	MBAI	86.58	87.84	81.37	3733958	5117076	516663	855091	84.48	73.13
16	NBAI	85.27	82.01	87.18	1353078	1517411	296811	198909	84.52	73.19
17	NBI	83.90	81.83	79.08	4678288	7186280	1038724	1237657	80.43	67.27
18	NDBI	82.46	87.75	82.44	3915715	2576427	546834	833981	85.01	73.93
19	PISI	82.97	82.98	95.94	6457849	1329437	1324984	272976	88.99	80.16
20	RNDSI	79.90	89.20	78.71	2287896	1273715	277039	619000	83.62	71.86
21	SwiRed	78.96	88.62	79.20	2511785	1172785	322447	659541	83.65	71.89
22	UBI	78.28	88.79	80.46	3184598	1051454	402000	773394	84.42	73.04
23	UI	86.23	87.46	85.26	5060467	4963479	725446	874949	86.35	75.97
24	VIBI	77.73	86.43	83.07	1597561	414265	250879	325553	84.72	73.49
25	WE-NDBI	79.19	82.63	91.63	5337123	788813	1121892	487766	86.90	76.83

Supplementary Table S13: Pixel level segmentation results on the SeasonNet dataset for the winter images. Each row represents one multispectral index. Accuracy, precision, recall, F1 score, and IoU are reported in percent. TP, TN, FP, and FN are total pixel counts summed over the complete winter split.

No.	Index	Acc (%)	Prec (%)	Rec (%)	TP	TN	FP	FN	F1 score (%)	IoU (%)
1	VrNIR-BI	83.68	85.85	81.54	52694097	53010744	8681890	11930917	83.64	71.88
2	VgNIR-BI	82.94	85.72	81.36	45468798	42003455	7577078	10420201	83.48	71.64
3	BBI	82.10	83.59	89.95	20868211	8614436	4097983	2331272	86.65	76.45
4	CBCI	83.02	84.88	83.81	66275848	54052537	11806313	12805678	84.34	72.92
5	BAEI	84.24	88.27	78.04	28204040	34223678	3746496	7935659	82.84	70.71
6	BCI	82.81	84.97	83.88	58896088	45816130	10416523	11319422	84.42	73.04
7	BLFEI	83.50	88.16	79.62	46535413	45385362	6249648	11912505	83.67	71.93
8	BRBA	84.86	88.35	78.00	22606709	29863377	2981392	6377857	82.85	70.72
9	BUB	81.78	84.83	81.91	50150175	39862546	8971314	11078998	83.34	71.44
10	BUC	82.40	86.49	82.68	41767161	29743703	6522913	8750628	84.54	73.22
11	BUI	84.04	88.39	79.42	47617262	50300403	6253542	12341596	83.66	71.92
12	CBI	83.51	86.20	80.32	36376308	38250676	5822234	8911255	83.16	71.17
13	ENDISI	80.30	86.47	78.19	25176368	19519082	3940574	7021456	82.12	69.67
14a	IBI with NDVI	81.94	84.20	84.40	65587262	42138530	12384220	12142479	84.30	72.80
14b	IBI with SAVI	82.01	84.42	83.90	65309028	46772006	12051766	12536034	84.16	72.65
15	MBAI	81.71	84.76	80.54	41410956	36462893	7443759	10004381	82.60	70.36
16	NBAI	83.36	86.30	79.59	21026486	22701969	3337614	5390387	82.81	70.67
17	NBI	80.93	81.88	72.59	38678367	59602022	8560355	14604905	76.96	62.54
18	NDBI	80.95	83.94	84.30	63058442	38074682	12061175	11744423	84.12	72.59
19	PISI	82.59	86.46	85.77	46210542	24494499	7237896	7668300	86.11	75.61
20	RNSDI	82.01	85.39	82.77	50084485	36559397	8572467	10428270	84.06	72.50
21	SwiRed	76.25	86.16	76.06	11781734	6200967	1893139	3707463	80.80	67.78
22	UBI	77.06	85.51	80.57	29141615	11054913	4938931	7029466	82.96	70.89
23	UI	81.61	83.29	86.28	77915719	46458568	15634874	12389312	84.76	73.55
24	VIBI	73.89	86.89	79.88	16829538	2345861	2539346	4237738	83.24	71.29
25	WE-NDBI	82.02	83.17	89.37	93524331	43540342	18929560	11125565	86.16	75.68

Supplementary Table S14: Pixel level segmentation results on the SeasonNet dataset for the spring images. Each row represents one multispectral index. Accuracy, precision, recall, F1 score, and IoU are reported in percent. TP, TN, FP, and FN are total pixel counts summed over the complete spring split.

No.	Index	Acc (%)	Prec (%)	Rec (%)	TP	TN	FP	FN	F1 score (%)	IoU (%)
1	VrNIR-BI	85.49	86.86	80.77	116029173	150150988	17549087	27633293	83.70	71.97
2	VgNIR-BI	84.77	85.99	82.05	113921498	128087392	18568084	24923483	83.97	72.37
3	BBI	84.69	95.77	73.00	484731	625977	21418	179295	82.85	70.72
4	CBCI	84.28	87.39	80.78	142896283	150021145	20617734	34000367	83.96	72.35
5	BAEI	85.87	88.40	78.93	97030553	137807525	12726784	25902119	83.40	71.53
6	BCI	84.24	87.32	80.52	109754420	117399222	15942911	26552795	83.78	72.09
7	BLFEI	84.67	87.36	81.05	129713771	141465559	18764946	30320357	84.09	72.55
8	BRBA	86.19	88.10	78.51	78368168	121535375	10589106	21447722	83.03	70.98
9	BUb	82.16	85.33	80.39	76048482	69581075	13075299	18551637	82.79	70.63
10	BUc	83.26	86.84	82.55	98747973	79502200	14968809	20869106	84.64	73.37
11	BUI	85.04	87.68	80.64	127750741	148571622	17949368	30678148	84.01	72.43
12	CBI	87.38	87.06	78.38	32444994	62880371	4821559	8951337	82.49	70.20
13	ENDISI	82.58	87.22	79.17	99747232	94039512	14619905	26244847	83.00	70.94
14a	IBI with NDVI	82.01	84.71	83.41	125718305	91765004	22684926	25006865	84.06	72.50
14b	IBI with SAVI	82.39	84.74	83.07	120342645	95715434	21667587	24525638	83.90	72.26
15	MBAI	82.70	86.59	79.20	90580633	90263149	14026642	23793293	82.73	70.55
16	NBAI	84.98	86.05	82.10	88356746	101691545	14325698	19258295	84.03	72.46
17	NBI	82.75	83.02	73.69	96312462	163246136	19703879	34389411	78.07	64.04
18	NDBI	82.38	83.85	85.80	144527046	97493024	27837184	23913187	84.82	73.63
19	PISI	85.01	88.04	80.37	24490714	28277188	3326092	5980169	84.03	72.46
20	RNDSI	82.36	87.48	80.43	100580123	80965166	14398720	24476423	83.80	72.12
21	SwiRed	73.86	88.13	74.14	11741682	4300945	1581265	4095967	80.53	67.41
22	UBI	78.61	86.02	82.00	77215686	31227755	12544579	16955449	83.96	72.36
23	UI	83.19	85.11	84.13	147990894	118329809	25898520	27913302	84.62	73.33
24	VIBI	73.48	88.03	78.45	20963996	2900013	2850979	5759776	82.96	70.88
25	WE-NDBI	83.97	86.17	84.58	177558819	138876817	28501772	32363221	85.37	74.47

Supplementary Table S15: Pixel level segmentation results on the SeasonNet dataset for the summer images. Each row represents one multispectral index. Accuracy, precision, recall, F1 score, and IoU are reported in percent. TP, TN, FP, and FN are total pixel counts summed over the complete summer split.

No.	Index	Acc (%)	Prec (%)	Rec (%)	TP	TN	FP	FN	F1 score (%)	IoU (%)
1	VrNIR-BI	84.70	86.82	80.29	103996881	124716551	15791531	25525178	83.43	71.57
2	VgNIR-BI	83.62	85.91	81.47	101309704	101101092	16618524	23040748	83.63	71.87
3	BBI	83.76	94.95	73.92	432989	473684	23044	152728	83.13	71.13
4	CBCI	84.06	87.49	80.18	131613152	139242181	18813153	32534761	83.68	71.94
5	BAEI	85.79	88.57	78.84	93043283	130218473	12006933	24976814	83.42	71.56
6	BCI	84.50	87.76	79.97	100971054	113650697	14087728	25291821	83.68	71.94
7	BLFEI	84.57	87.65	81.13	132021927	138192128	18606689	30704699	84.26	72.81
8	BRBA	86.33	88.46	78.45	76540655	119315546	9987912	21026683	83.15	71.16
9	BUb	82.19	85.44	80.17	74755983	69350712	12736500	18485140	82.73	70.54
10	BUc	82.05	87.30	80.62	75663550	57805435	11007576	18189737	83.83	72.16
11	BUI	85.13	88.06	80.58	127184232	147095688	17249732	30642213	84.16	72.65
12	CBI	87.01	86.82	78.72	32072449	58595888	4869149	8672185	82.57	70.31
13	ENDISI	81.65	87.05	79.08	87659442	73508549	13036830	23188351	82.88	70.76
14a	IBI with NDVI	81.31	84.25	83.29	113045281	77619164	21139683	22677993	83.77	72.07
14b	IBI with SAVI	81.80	84.35	82.96	108654071	82247720	20157368	22312762	83.65	71.90
15	MBAI	81.85	86.38	78.79	77822776	72009938	12270233	20951435	82.41	70.08
16	NBAI	84.54	86.02	81.63	84435966	94616705	13726904	19005882	83.76	72.06
17	NBI	81.73	82.17	73.37	85421157	136131111	18540413	31001119	77.52	63.29
18	NDBI	81.25	83.76	84.86	128457149	78744583	24905907	22922859	84.31	72.87
19	PISI	84.27	88.24	79.95	21122950	22338403	2814140	5296692	83.89	72.26
20	RNDSI	82.35	86.04	81.50	107697426	87832170	17477315	24443037	83.71	71.98
21	SwiRed	74.50	87.17	74.88	11838319	4853221	1741925	3970521	80.56	67.45
22	UBI	79.09	85.82	82.19	86128577	38285864	14232364	18663456	83.97	72.36
23	UI	82.33	84.87	83.75	139974604	102791894	24948743	27169222	84.30	72.87
24	VIBI	74.58	87.07	78.85	23170569	5162933	3441748	6213344	82.76	70.59
25	WE-NDBI	84.10	86.48	84.20	175192898	143754587	27400556	32882847	85.32	74.40